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Transfer of the high-temperature adult-plant stripe rust resistance gene *Yr62* in four Chinese wheat cultivars

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Abstract Wheat stripe rust is one of the diseases that seriously affect wheat production worldwide. Breeding resistant cultivars is an effective way to control this disease. The wheat stripe rust resistance gene *Yr62* has high-temperature adult-plant resistance (HTAP). In this study, PI 660,060, a single *Yr62* gene line, was crossed with four Chinese wheat cultivars, LunXuan987 (LX987), Bainongaikang58 (AK58), ZhengMai9023 (ZM9023), and HanMai6172 (H6172). F_1 seeds of four cross combinations were planted and self-crossed to develop the advance generations in the field. The seeds of each cross were mixed harvested and about 2400 to 3000 seeds were sown in each generation for F_1 to F_4 to maintain the maximum possible genotypes. Forty-five lines were selected and evaluated for resistance to stripe rust and agronomic traits, including plant height, number of grains per spike, and tiller number, in F_5 and F_6. Then, 33 lines with good agronomic traits and high disease resistance were developed to F_9 generation. SSR markers *Xgwm251* and *Xgwm192* flank linked with the *Yr62* were used to detect the presence of *Yr62* in these 33 F_9 lines. Of these, 22 lines were confirmed with the resistance gene *Yr62*. Finally, nine lines with good agronomic traits and disease resistance were successfully selected. The selected wheat lines in this study provide material support for the future breeding of wheat for stripe rust resistance.

Keywords Wheat · Stripe rust · HTAP resistance · *Yr62* · Disease resistance breeding · SSR maker detection

Introduction

Stripe (yellow) rust of wheat, caused by *Puccinia striiformis* f. sp. *tritici* (*Pst*), is one of the most important wheat diseases in the world and *Pst* can

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significantly reduce wheat yield and quality (Roelfs and Bushnell 1985). By 2050, global wheat production will need to increase by 60% to feed 9.6 billion people worldwide, given the challenges posed by diminishing land and water resources and changing climate conditions (Savadi 2018). Yield losses due to stripe rust vary from 10 to 70%, depending on prevailing weather conditions, cultivars, and disease development (Chen 2005b; Zhou et al. 2021). Rising global temperatures may affect the growth stage of wheat, accelerate the senescence of wheat leaves, and thus affect the breeding of stripe rust, or even directly block it. However, stripe rust also evolves with increasing temperature and adaption to high-temperature environments, so the introduction of the high-temperature adult-plant resistance gene is beneficial to enhancing wheat resistance (Asseng et al. 2011; Chakraborty and Newton 2011; Juroszek and von Tiedemann 2013; Khanfri et al. 2018). *Pst* is suitable for survival at low temperatures. Some areas in northwest Sichuan, China have high altitude and low temperature, and wheat is vertically planted at different altitudes, which leads to the successful completion of oversummering, overwintering, and the annual life cycle of stripe rust. Such an environment is not only conducive to the generation new stripe rust races, but also suitable for the occurrence of wheat stripe rust in the regions (Chen 2005a; Li et al. 1997; Liu and Shen 2002; Xiang et al. 2013). The most effective, economical, and environmentally friendly way to control wheat stripe rust is to plant resistant cultivars (Line and Chen 1995; Parlevliet 1985). Wheat stripe rust resistance can be divided into all-stage resistance (ASR) and adult-plant resistance (APR). APR genes often confer incomplete resistance and have historically been more durable than race-specific genes. Therefore, a promising approach to the problem of loss of disease resistance yield is the use APR genes, which are effective in the adult plant stage and are considered to have a broader range of resistance and longer durability (Brown 2015; Lowe et al. 2011; Niks et al. 2015). The most important type of APR in wheat is high-temperature (10−30℃) adult-plant resistance (HTAP). HTAP resistance is non-race specific and durable (Chen 2005b, 2013; Line 2002), and its resistance level gradually increases with the plant growing to adult stage and the environmental temperature increasing. Therefore, breeding HTAP cultivar is one of the most promising methods to control wheat stripe rust. Conventional breeding, however, is a time-consuming and labor-intensive process that can take 7 to 20 years. Marker-assisted selection (MAS) is a combination of modern molecular biology and conventional breeding, which selects breeding wheat lines at the DNA level by employing molecular markers, to achieve efficient improvement of comprehensive traits such as crop yield, quality, and disease resistance (Collard et al. 2005). Currently, MAS breeding is the most effective and rapid method for selecting excellent plants containing target genes. Since 2000, simple sequence repeat (SSR) markers have gradually become widely used in resistance gene mapping and in introducing resistance genes into wheat cultivars (Akbari et al. 2006). For example, *Yr76* was identified with flanking SSR markers *WMC11* and *WMC532* in a F_2 mapping population of Tyee and Avocet Susceptible (AvS) crosses (Xiang et al. 2016). Three SSR markers, *Xics13*, *Xics43*, and *Xics76*, were highly polymorphic among the different wheat cultivars used for validation, which were added to the *Pm4b* genetic linkage map (Wu et al. 2018). Christiansen et al. reported a yellow rust resistance QTL in the population of a cross between Deben and Wasmo on chromosome 6BL linked to SSR markers *Xwmc397* and *Xwmc105b* (Christiansen et al. 2006). *QYr.cimmyt-6BL* was mapped to 6BL with SSR markers *Xgwm58* and *Xgwm626* in Pavon76 (William et al. 2006). Four stem rust resistance genes (*Sr33*, *Sr36*, *Sr-Cad*, and *Sr43*) and three stripe rust resistance genes (*Yr5*, *Yr18*, and *Yr26*) were simultaneously infiltrated into Plateau448 by marker-assisted backcross hybridization combined with phenotypic selection, which improved stem rust resistance (Ug99) and stripe rust (Liu et al. 2020).

Yr62 is the HTAP resistance gene and was identified from Portuguese spring wheat cultivar PI 192,252, which was closely linked to two SSR markers *Xgwm251* and *Xgwm192* (Lu et al. 2014). PI 660,060 is a progeny line selected from cross AvS and PI 192,252 and carrying *Yr62* with relatively good agronomic traits, and has maintained good resistance to stripe rust in multiple experimental sites in the Washington State of the United States for many years (Wang et al. 2012b). PI 660,060 also showed good resistance to stripe rust in several provinces of China over a 5-year field test (Zhou et al. 2015). To improve wheat resistance to stripe rust in China, it is

better to introduce HTAP resistance, such as *Yr62*, into local cultivars.

The purposes of this study are to (1) screen the phenotypes of the progeny of the breeding population, and then use marker selection to transform *Yr62* into four excellent Chinese wheat cultivars, (2) screen new wheat germplasm with excellent agronomic traits and strong resistance to stripe rust, and (3) speed up the molecular design breeding of wheat resistance to stripe rust. It provides a scientific basis for further research on the utilization and breeding of resistance genes.

Materials and methods

Plant materials

PI 660,060, a single gene line containing the HTAP resistance gene *Yr62*, was used as the resistant donor parent. Four elite Chinese cultivars LX987, AK58, ZM9023, and H6172 were used as recipient parents in this study.

LX987 is a winter wheat variety bred through recurrent selection breeding of dwarf and sterile wheat and pedigree method. It has excellent quality and disease resistance and has been widely planted in the Huang Huai wheat region (Wang et al. 2010; Yang 2006). AK58 is a semi-winter wheat cultivar bred from Zhoumai11//Wenmai6/Zhengzhou8960. It has the characteristics of high yield, lodging resistance, disease resistance, and stable yield (Wang et al. 2012a). ZM9023, a weak spring cultivar developed from the cross of Shan213//Xiaoyan6/Xinong65, has high salt tolerance, superior gluten quality, and resistance to Fusarium head blight (*QFhb.7D*) (Gao et al. 2011; Xiang et al. 2016). H6172 is a semi-winter wheat developed from the 4032/Zhongyin1 cross. It has excellent characteristics such as strong adaptability, drought resistance, and immunity to stem rust (Sun et al. 2004). These four cultivars have high yields, with the exception of H6172, which has a slow rust resistance. The other three cultivars have strong resistance to stripe rust when released. However, they are not effective to the current stripe rust race CYR34 (Liu et al. 2017). However, due to their special agronomic properties, they can continue to be used as recipient parents. Spring wheat germplasm PI 192,252 showed a high level of HTAP resistance to stripe rust in germplasm evaluation from to 2014 in the State of Washington (Lu et al. 2014; Wang et al. 2012b). In Yangling city (Shaanxi province), Tianshui city (Gansu province), and Mianyang (MY) city (Sichuan province), China, PI 660,060 also maintained high levels of resistance to stripe rust [infection type (IT) 2–3, disease severity (DS) 2–30%] from 2010 to 2014 (Zhou et al. 2015). PI 660,060 had the same resistance to stripe rust and relatively better agronomic traits compared to PI 192,252 (Wang et al. 2012b) and was used as *Yr62* donor parent in crosses with four Chinese cultivars.

Construction breeding populations and screening of recombinant inbred lines

PI 660,060 was crossed with LX987, AK58, ZM9023, and H6172, of which PI 660,060 was the donor parent and four superior cultivars were recipients. The harvested F_1 seeds were sown in the field and selfed to develop the advance generations. The F_2–F_5 seeds of each hybrid combination were sown in 30 rows, 2 m long and 25 cm apart, with approximately 80–100 seeds in each row. In generation F_4, progeny lines were selected based on the following criteria: plant height (PH, cm) < 100, number of grains per spike (SGN) ≥ 40, tiller number (TN) ≥ 4, and IT < 6. Before the F_5 generation, we used a mixed harvesting method for each combination to retain all possible genotypes. The selected lines were increased to F_6 generation in the field. In F_6, lines were again selected according to the criteria: 80 < PH < 90, SGN ≥ 40, TN ≥ 4, thousand grain weight (TGW, g) ≥ 40, and IT 0–5. Selected lines continue to be sown and evaluated as previously described and added to the F_9 generation. Finally, we obtained selected F_9 lines with stable and excellent resistance to stripe rust, moderate plant height, high number of tillers and spikelets, and high yield (Fig. 1).

Disease resistance and agronomic trait evaluation in the field

In the growing seasons of 2018–2022, parent lines PI 660,060, LX987, AK58, ZM9023, H6172, and the F_6 to F_9 selected progeny lines were evaluated for stripe rust resistance and agronomic traits at the experimental site in MY, Sichuan (31°33′N, 104°55′E, 485 m above sea level). Each line was sown in 3 rows, with

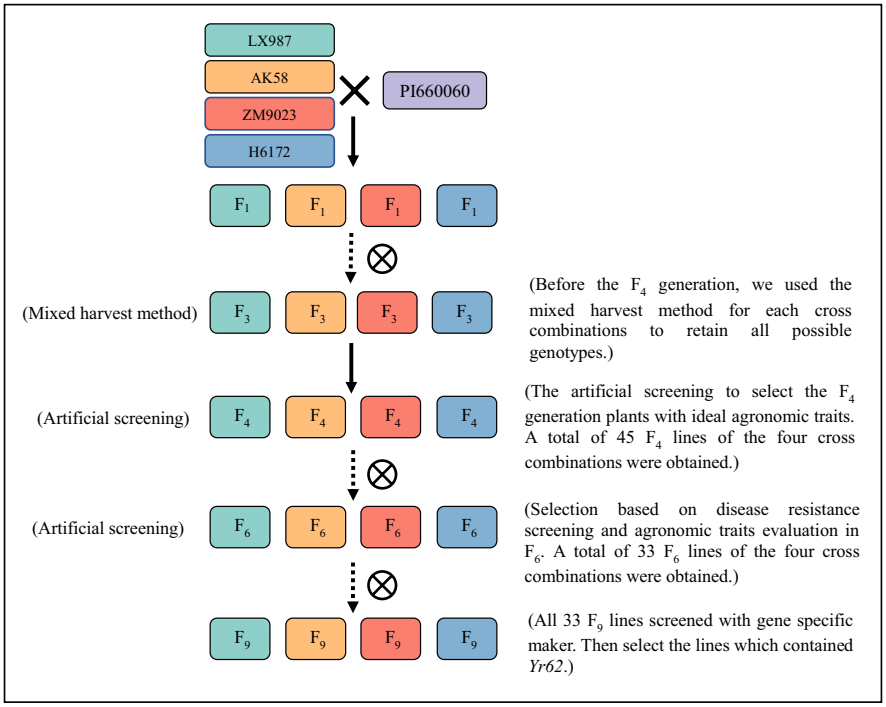

Fig. 1 The scheme of development, validation, and selection of a wheat line approach for introgression of *Yr62* stripe rust resistance gene into four elite cultivars LX987, AK58, ZM9023, and H6172 from PI 660,060(*Yr62*) resistant donor

a row length of 1 m, a row spacing of 25 cm, and about 30 seeds in each row. To increase field inoculum for stripe rust, susceptible cultivar Mingxian169 (MX169) was planted around the experimental plots.

Stripe rust on each row was evaluated from heading (Z50) to grain filling (Z80) (Zadoks et al. 1974). Under the condition that the severity of stripe rust in the susceptible wheat variety MX169 was 80–100%, the infection type (IT) and disease severity (DS) of 5 parents and their selected lines were recorded twice respectively. IT was recorded following the 0–9 scale (Line and Chen 1995), with 0 considered immune, 1–3 highly resistant, 4–6 moderately resistant, and 7–9 susceptible. DS was recorded as 0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, or 100 for the percentage of leaf area infected by *Pst*. The survey results are average.

Agronomic traits were measured according to the following criteria: Plant height (PH, cm): the height from the tiller node to the top of the spike (excluding awns), taking the average height of 10–20 plants. Spike length (SL, cm): length from the base to the top of the spike (excluding awns). Number of grains per spike (SGN): the total number of grains on the spike of wheat. Thousand grain weight (TGW, g): take dry and thoroughly dried grain as the standard. After mixing the samples, randomly take 1000 grains and weigh them with a 1/100 electronic balance. Tiller number (TN): the number of tiller which produce normal spike and bearing grains. All screened lines were named with the abbreviation following the nomenclature system of Southwest University of Science and Technology (SWUST).

Genotyping wheat lines by markers

In 2022, fresh leaves of F_9 lines and 5 parents were collected in the MY experimental field, and DNA was extracted by the modified cetyltrimethylammonium bromide (CTAB) method (Cuc et al. 2008). The parents and selected progeny lines were genotyped using two SSR markers, *Xgwm192* and *Xgwm251*. Select lines with the same bands as donor parent PI 660,060 for both markers indicated that *Yr62* was successfully transferred to progeny lines.

The polymerase chain reaction (PCR) system is as follows: the total volume of the system is 10 µl, which contains 1 µl of 10×PCR buffer, 0.2 µl of Taq enzyme, 0.8 µl of dNTP, 4 µl of dd H_2O, forward and reverse primers in a volume of 1 µl each, and 2 µl of DNA sample (the concentration of all DNA samples is between 50 ng/µl). The sequences of the two

Table 1 Sequence and amplification information for SSR markers linked to *Yr62*

Marker	Primer sequence (5′–3′)	T_m (℃)	Reaction procedure	References
Xgwm192	F:GGTTTTCTTTCAGATTGCGC R:CGTTGTCTAATCTTGCCTTGC	55.6	94℃ 4 min; 35 cycles (94℃ 30 s; 57℃ 30 s; 72℃ 30 s); 72℃ 8 min; 16℃, ∞	(Lu et al. 2014)
Xgwm251	F:CAACTGGTTGCTACACAAGCA R:GGGATGTCTGTTCCATCTTAG	54.5	94℃ 4 min; 35 cycles (94℃ 30 s; 57℃ 30 s; 72℃ 30 s); 72℃ 8 min; 16℃, ∞	(Lu et al. 2014)

primers used in the molecular marker test and the reaction procedures are shown in Table 1. The PCR amplification products were loaded with 6 µl of loading buffer [98% formamide, 10 mM EDTA (pH 8.0), 0.5% w/v xylene/cyanol, and 0.5% w/v bromophenol blue]. For each sample, 4 µl of PCR products and loading buffer mixture were separated by 6% polyacrylamide gel electrophoresis (PAGE) after denaturation at 94 °C for 4 min. After electrophoresis, the plate was removed and placed horizontally in a water basin to be rinsed with distilled water, then silver nitrate solution ($AgNO_3$: 25 g, distilled water: 2.5 ml) for 30 min for light staining; after silver staining, the plate was placed in distilled water for 5–10 s, then a mixture of sodium hydroxide and formaldehyde solution (NaOH: 25 g, HCHO: 4 ml) was added for development, shaken until the bands were clear, the plate was removed and placed in distilled water to clean it, dried, and then placed on the scanner to be observed (Chen et al. 1998; Zhou et al. 2014) (Supplementary Fig. 1).

Statistical analysis

Variance, correlation, and broad-sense heritability of the phenotype data of progeny lines were calculated using software IciMapping V4.2 (Meng et al. 2015). Broad-sense heritability (h^2) of phenotypic traits was calculated using the formula $h^2 = \sigma_g^2/(\sigma_g^2 + \sigma_{ge}^2/r + \sigma_\varepsilon^2/re)$, where σ_g^2, σ_{ge}^2, and σ_ε^2 are estimates of genotypic, genotype×environment interaction, and error variances, respectively, and e and r are the number of environments and replications per environment, respectively (Meng et al. 2015; Wang 2009).

To compare the effect of *Yr62* on TGW and SGN, we sorted progeny lines into two groups, one group having *Yr62* and another without *Yr62*. The difference between two groups were determined by *t* test in software GraphPad Prism version 8.0.2.

Results

Initial selection for stripe rust resistance and Yr62 molecular marker detection

The four main Chinese cultivars and PI 660,060 were hybridized, self-crossed, and screened according to the method shown in Fig. 1. Because the climate in Sichuan Basin is suitable for wheat stripe rust, the disease occurred in all test fields in MY every crop season. F_4 plants were screened according to the agronomic characters and stripe rust resistance, and 45 F_4 plants were obtained. Then, 45 F_5 seeds were sown in the field to complete the self-crossing. Thirty-three plants were obtained based on agronomic traits and stripe rust resistance criteria. Thirty-three F_7 generation lines were screened again. F_8 generation seeds were sown in the field for self-crossing to F_9 generation, and 33 F_9-generation lines with excellent agronomic traits and stable disease resistance were obtained.

The 33 F_9 lines were then tested with *Xgwm192* and *Xgwm251*. The result showed that 22 lines (67%) contained *Yr62*, and 11 lines did not have *Yr62* (Supplementary Fig. 1). Of these 22 lines, two lines were from LX987/PI 660,060, two lines were from AK58/PI 660,060, five lines were from ZM9023/PI 660,060, and thirteen lines were from H6172/PI 660,060 (Table 2).

Evaluation of disease resistance

Based on field disease resistance identification, recipient parents LX987, AK58, ZM9023, and H6172 showed intermediate resistance to

Table 2 The number lines of presence (+) or absence (−) of *Yr62* in 33 selected F_9 progeny lines from 4 crosses detected with SSR markers *Xgwm192* and *Xgwm251*

Cross combination	*Yr62*(+)	*Yr62*(−)
LX987/PI 660,060	2	2
AK58/PI 660,060	2	1
ZM9023/PI 660,060	5	1
H6172/PI 660,060	13	7

susceptibility to stripe rust at the adult stage in the field, and the 4-year mean IT was 4, 7, 5, and 6, respectively (Tables 3). DS was 20%, 55%, 50%, and 35%, respectively. The resistance donor PI 660,060 (*Yr62*) showed high resistance (IT 1, DS 10) at the adult stage in the field. The disease resistance of progeny with the *Yr62* gene is shown in Fig. 2. The 33 advanced generation lines from the four crosses had varied IT and DS to stripe rust (Fig. 3, Table 3). The IT of 33 higher generation lines (Table 3, Fig. 3) was mainly distributed in 1–6, and no susceptible and immune lines were found. The DS is mainly distributed at 5–50% and is less than 40% (Fig. 3). Of the four lines from cross LX987/PI 660,060, SWUST-413 was highly resistant to intermediately resistant (IT 0, 4, 3, and 5; DS 0, 25%, 8%, and 30% in 2019, 2020, 2021, and 2022, respectively), and SWUST-407, SWUST-408, and SWUST-409 were intermediately resistant. The disease resistance of the three lines of AK58/PI 660,060 was higher than that of the recipient parent. SWUST-419 reached the high resistance level (IT 4, 4, 3, and 3; DS 8%, 30%, 20%, and 13%), and SWUST-416 and SWUST-420 reached the intermediate resistance level. Among the six progeny lines of ZM9023/PI 660,060, SWUST-421, SWUST-425, SWUST-427, and SWUST-428 reached high resistance levels, and SWUST-423 and SWUST-426 reached intermediate resistance levels. Among the 20 progeny lines of H6172/PI 660,060, SWUST-441 (IT 0, 2, 2, 2; DS 0, 10, 6, 6) and SWUST-447 (IT 0, 1, 1, 2; DS 0, 3, 3, 6) both reached the same resistant level with *Yr62* (IT 1, 1, 1, 0; DS 10, 15, 10, 0); the other progeny lines SWUST-431, SWUST-436, SWUST-437, SWUST-438, SWUST-439, SWUST-442, SWUST-443, SWUST-444, SWUST-448, and SWUST-449 reached the high resistance level; and SWUST-430, SWUST-432, SWUST-433, SWUST-434, SWUST-435, SWUST-440, SWUST-450, and SWUST-451 were in intermediate resistance level.

The correlation analysis of disease resistance phenotypes of each family in different years showed that the correlation between IT and DS of each family in 2019 and 2020 was extremely significant ($P<0.001$), and the correlation coefficients were $r=0.78$ and $r=0.84$, respectively (Table 4). There was no significant difference in IT and DS correlation between 2019 and 2021, 2020 and 2021, and 2021 and 2022 ($r=0.11$–0.36, Table 4). The broad-sense heritability of IT and DS is 0.67 and 0.75, respectively. An ANOVA of IT and DS revealed significant differences ($P<0.01$) in four environments, but not significant in line × environment interactions. This shows that each family is greatly affected by the environment due to the great changes in races every year (Table 5).

Evaluation of agronomic traits

In the 2021–2022 crop seasons, the mean plant height of the 5 parents, *Yr62*, LX987, AK58, ZM9023, and H6172, were 88, 84, 77, 86, and 88 cm, respectively (Table 3, Fig. 4). Among the progeny lines, PH values were mainly in the 80–120 cm range. The mean SGN values of the 5 parents were 20, 49, 39, 55, and 57. Of the 33 lines, 32 lines had SGN values ≥ 45. The mean tiller values of the five parents were 9, 6, 5, 5, and 5, respectively. All 33 lines had TN values ≥ 5. The mean SL values of the 5 parents were 11, 13, 12, 12, and 14. Also, the SL value of all progeny lines is between 12 and 16. The mean TGW of the five parents was 16, 47, 46, 44, and 50 g, respectively. The average TGW values of the four combinations were mainly between 40 and 55 g (Fig. 4, Table 3). By comparing the TGW and SGN of progeny lines with or without *Yr62*, the mean TGW of more than half of the 33 lines was in the range of recipient parents (47–50 g), and similarly, the mean SGN of more than half of the 33 lines was in the range of recipient parents (45–60 grains) (Fig. 5).

Discussion

Evaluation of wheat stripe rust

Wheat stripe rust is a major fungal disease that seriously threatens the safety of wheat production. It

Table 3 The average values of resistance to stripe rust, agronomic traits, and molecular marker associated with *Yr62* in the wheat parents and selected progeny lines tested during 2019–2022 seasons

Parent/line	Cross	Stripe rust		Agronomic trait					*Yr62*(+/−)
		IT	DS (%)	PH (cm)	SGN	TN	SL (cm)	TGW (g)	
PI 660,060		1	10	88	20	9	11	16	+
LX987		4	20	84	49	6	13	47	−
AK58		7	55	77	39	5	12	46	−
ZM9023		5	50	86	55	5	12	44	−
H6172		6	35	88	57	5	14	50	−
SWUST-407[a]	LX987/PI 660,060	4	10	85	78	5	16	46	+
SWUST-408	LX987/PI 660,060	4	40	111	54	8	14	45	+
SWUST-409	LX987/PI 660,060	4	20	80	54	9	12	48	−
SWUST-413	LX987/PI 660,060	3	20	88	54	7	13	43	−
SWUST-416	AK58/PI 660,060	4	30	80	51	6	13	50	+
SWUST-419	AK58/PI 660,060	3	10	111	59	8	15	47	−
SWUST-420[a]	AK58/PI 660,060	4	40	105	58	6	15	45	+
SWUST-421	ZM9023/PI 660,060	3	30	82	51	9	14	43	+
SWUST-423	ZM9023/PI 660,060	4	30	97	52	6	14	46	+
SWUST-425	ZM9023/PI 660,060	3	20	81	40	7	13	53	+
SWUST-426[a]	ZM9023/PI 660,060	4	30	92	54	7	14	50	+
SWUST-427[a]	ZM9023/PI 660,060	3	20	96	49	6	13	50	−
SWUST-428	ZM9023/PI 660,060	3	20	90	55	9	14	41	+
SWUST-430	H6172/PI 660,060	4	30	94	48	6	14	43	−
SWUST-431	H6172/PI 660,060	3	10	102	46	8	12	42	−
SWUST-432	H6172/PI 660,060	5	20	105	55	6	13	47	−
SWUST-433	H6172/PI 660,060	4	20	93	48	5	13	44	+
SWUST-434	H6172/PI 660,060	4	20	98	52	6	14	50	+
SWUST-435	H6172/PI 660,060	4	20	97	52	7	14	44	+
SWUST-436[a]	H6172/PI 660,060	3	10	85	64	6	14	51	+
SWUST-437[a]	H6172/PI 660,060	3	10	89	59	6	15	52	+
SWUST-438[a]	H6172/PI 660,060	3	10	87	61	5	15	51	+
SWUST-439[a]	H6172/PI 660,060	2	10	87	64	5	16	50	+
SWUST-440	H6172/PI 660,060	4	20	86	60	6	15	50	+
SWUST-441	H6172/PI 660,060	1	5	112	61	7	15	56	+
SWUST-442	H6172/PI 660,060	2	10	91	49	8	15	45	+
SWUST-443	H6172/PI 660,060	3	20	90	52	8	14	40	−
SWUST-444[a]	H6172/PI 660,060	2	10	105	54	5	14	55	+
SWUST-447	H6172/PI 660,060	1	5	111	59	5	14	55	+
SWUST-448	H6172/PI 660,060	3	10	113	49	6	14	55	−
SWUST-449	H6172/PI 660,060	3	20	94	52	6	13	42	−
SWUST-450	H6172/PI 660,060	4	20	87	55	6	14	52	−
SWUST-451	H6172/PI 660,060	4	30	97	45	6	13	48	+

+ means that *Yr62* is detected, − means that *Yr62* is not detected

IT infection type, *DS* disease severity, *PH* plant height, *SGN* number of grains per spike, *TGW* thousand grain weight, *SL* spike length, *TN* tiller number

[a]Finally selected nine lines with excellent agronomic traits and high-level resistance to stripe rust

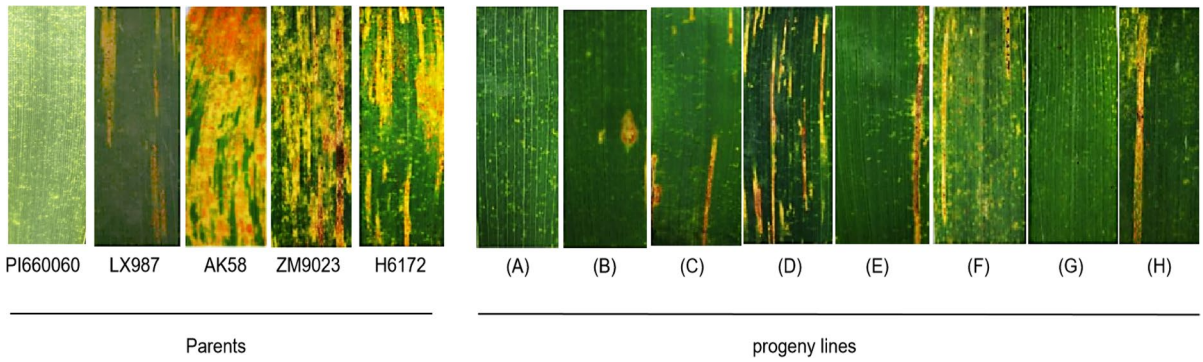

Fig. 2 Stripe rust reactions on leaves of four parents and the selected progeny lines. (A, B) Lines from LX987/PI 660,060, (C, D) lines from AK58/PI 660,060, (E, F) lines from ZM9023/PI 660,060, (G, H) lines from H6172/PI 660,060

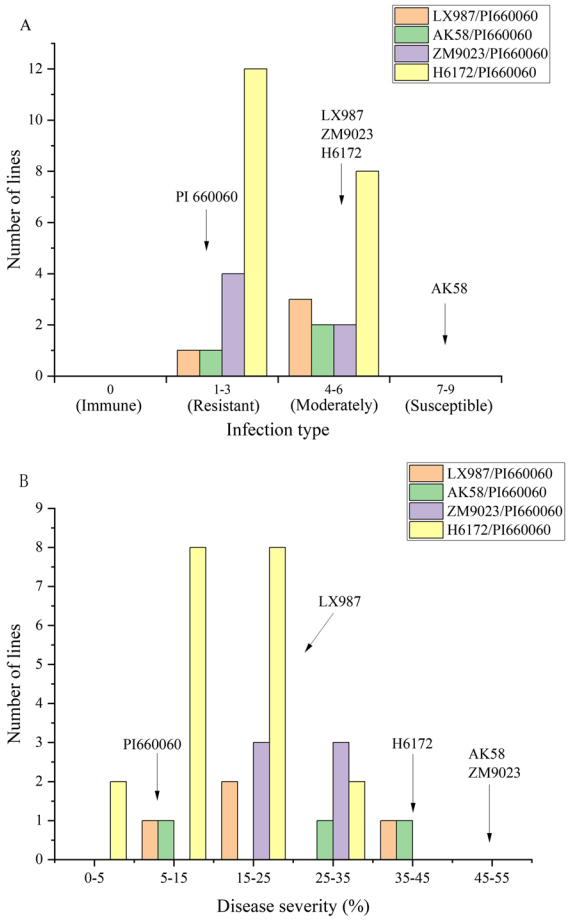

Fig. 3 Frequency distribution of infection type (A) and disease severity (B) of 33 wheat selected progeny lines from four crosses for resistance to stripe rust. The infection type and severity of each parent are indicated with arrows

has a wide distribution, strong epidemics, and serious harm, so it is an important factor restricting stable and high wheat yields (Guo et al. 2010). In the process of long-term prevention and control of stripe rust, Chinese and international scientific research workers have concluded that the use of resistant cultivars is the most economical, effective, and environmentally friendly measure to prevent and control the disease. However, due to the frequent occurrence of virulence variation in *Pst* populations, the occurrence of new *Pst* physiological races usually resulted in the major resistant varieties becoming susceptible, resulting in a loss of utilization value. Moreover, due to the special geographical location of Sichuan, China, it is also known as the "over summering variable area" of wheat stripe rust or the source of new stripe rust races (Kang et al. 2015). Therefore, breeding durable

Table 4 Correlation coefficients of infection type (IT) and disease severity (DS) of all wheat lines among 4 years

Resistant phenotypes	Year	2019	2020	2021	2022
IT	2019	1			
	2020	0.78***	1		
	2021	0.21 ns	0.21 ns	1	
	2022	0.36*	0.36*	0.11 ns	1
DS	2019	1			
	2020	0.84***	1		
	2021	0.29 ns	0.27 ns	1	
	2022	0.36*	0.36*	0.30 ns	1

Significance level at $P<0.05$ (*), $P<0.01$ (**), $P<0.001$ (***), not significant at $P>0.05$ (ns)

IT infection type, *DS* disease severity

Table 5 Results of ANOVA of infection type and disease severity of 33 recombinant inbred lines in 2019–2022

Source of variation	IT				DS (%)			
	df	MS	*F* value	[c]F. Sig	*df*	MS	*F* value	[c]F. Sig
Lines	32	5.97	3.36	***	32	620.06	4.31	***
Block/environments	4	36.16	20.36	***	4	2113.66	14.71	***
Environments	3	34.78	19.58	***	3	2596.46	18.07	***
Lines × environments	96	2.07	1.17	ns	96	159.34	1.11	ns
Error	128	1.78			128	143.72		
[a]σ^2_g	0.52				59.54			
[b]H^2	0.67				0.75			

[a]σ^2_g was estimated for genotypic (line) variances

[b]H^2 indicates the estimated heritability in the broad sense based on the mean across replications and environments (or heritability per mean)

[c]F. Sig: significance level at $P < 0.05$ (*), $P < 0.01$ (**), $P < 0.001$ (***)

disease-resistant cultivars is the most direct, economical, and effective method.

In this study, PI 660,060 containing *Yr62* showed excellent resistance to stripe rust in the adult stage (IT 1, DS 10%). Lu et al. at Pullman and Mt Vernon, two experimental sites with different races, identified PI 192,252 (*Yr62*) resistance to stripe rust in the field and resistance to different races at a high diurnal temperature cycle (10–30 °C) in the greenhouse at the adult stage. The results showed that PI 192,252 had typical HATP resistance (Lu et al. 2014). The four main cultivars LX987, H6172, AK58, and ZM9023 selected in this study showed intermediate resistance at the field mature stage (IT 4–7, DS 20–55%). Similarly, Xue used CYR32 and CYR33 to identify the resistance of 74 wheat cultivars at the adult stage for two consecutive years. Among them, research shows that under infection of two races, the IT of LX987, H6172, AK58, and ZM9023 are 8, 8, 5, 7 and 7, 7, 7, 8 in the seeding stage. Under the infection of mixed races in the adult stage, H6172 shows moderate resistance to moderate susceptibility, LX987 shows susceptibility, AK58 shows resistance to middle susceptibility, and ZM9023 shows susceptibility (Xue et al. 2014). Combined with this study, IT and DS were shown that infected plants of the same rust race from the same cultivar can vary in different environments (Rasheed et al. 2018). In this experiment, the IT and DS of four recipient parents in 2019 and 2022 are different. For example, the IT and DS of AK58 in 2019 and 2020 are 8, 65% and 3, 30%, respectively. The IT and DS of H6172 in 2020 and 2021 are 7, 50% and 3, 10%, respectively. The other two main cultivars LX987 and ZM9023 maintained intermediate resistance levels in 2019–2022 (Supplementary Table 1). The four main cultivars differ slightly in IT and DS each year or even vary dramatically between 2 years, and there are many reasons for this: (1) the race composition is different every year; (2) the expression of the high-temperature adult-plant gene is affected by temperature and/or plant development stage; (3) field temperature and inoculation method of stripe rust (Wellings 2011; Zeng et al. 2022). It is reported that ZM9023 contains APR genes *Yr29* and *Yr30* (Li et al. 2020). Therefore, in the progeny lines of the hybrid combination ZM9023/PI 660,060, SWUST-427 has not detected the *Yr62* gene, but it still maintains a high level of resistance to stripe rust. The lines without *Yr62* in the remaining three crosses also maintained similar stripe rust resistance with the recipient parents, possibly because AK58 contained *Yr9*, and LX987 and H6172 contained unknown resistance genes (Ma et al. 2021). The progeny lines that successfully introduced the *Yr62* gene maintained stable resistance to stripe rust in the field (IT 1–4, DS 5–40%) (Table 3).

Molecular marker evaluation

Over the past few decades, desirable traits in wheat have often been obtained through crossbreeding in traditional improvement programs, without knowledge of the use of heritable molecular mechanisms

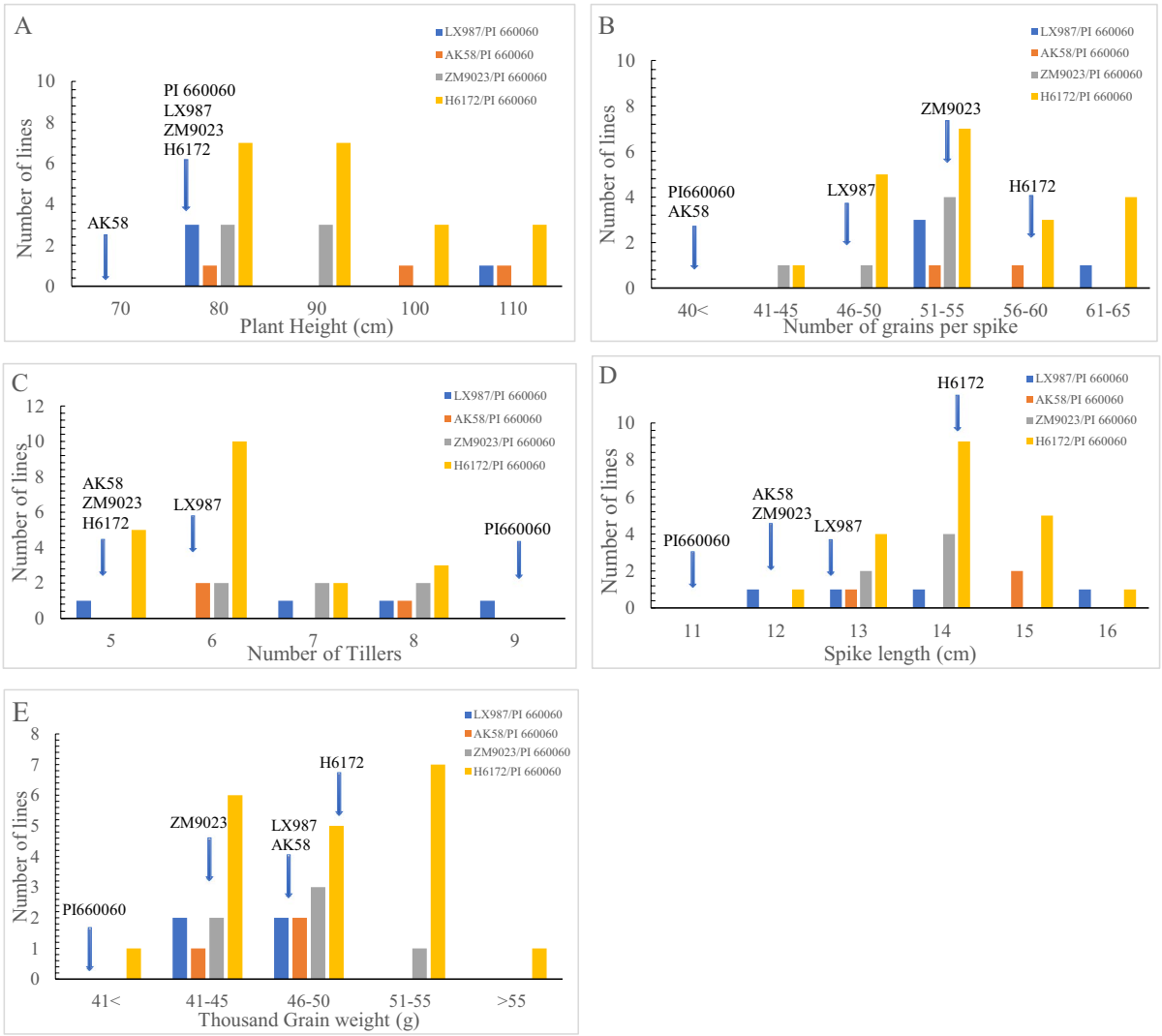

Fig. 4 **A–E** Frequency distribution of five agronomic traits of 33 progeny lines from four cross combinations

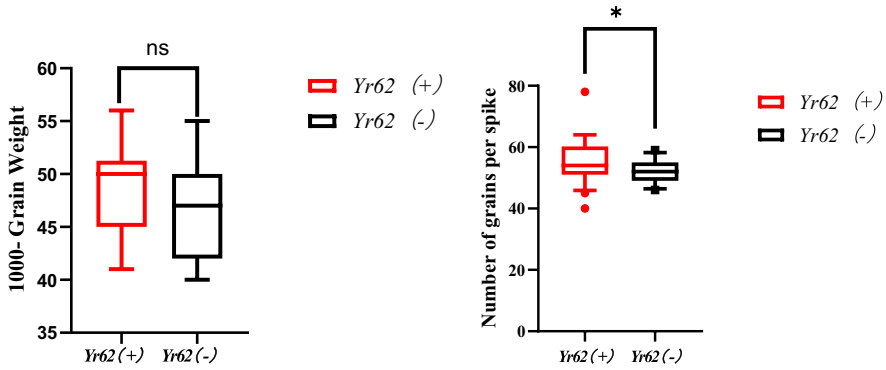

Fig. 5 Frequency distribution of thousand grain weight and grain per spike of 33 wheat progeny lines for present (+) or absent (−) resistance gene *Yr62* from four crosses. *p* value was generated by *t* test. "ns" represents $p > 0.05$, no significant differences; "*" represents $p < 0.05$, significant differences

(Susanne et al. 2016). This lengthy process preserves desirable traits in wheat, protects it from various disease stresses, and improves wheat yield and quality (Rasheed et al. 2018). As a complementary tool to traditional breeding, MAS accelerated the wheat breeding process. It can be used to select early target traits, especially when the shape is under complex genetic mechanisms or difficult to study. It can provide breeders with new ways to select traits and devote limited resources and energy to more promising crosses and progeny (Facundo et al. 2017). The molecular marker has been developed mainly including AFLP, RAPD, SCAR, RGAP, SSR, STS, and SNP. However, SSR is the most widely used molecular marker in the last 5 years. For example, *Yr59* was successfully introgressed into Jimai22 Chuanmai42, Zhengmai9023, and Xinmai26 by marker-assisted backcross selection and screened for 16 lines with agronomic traits similar or superior to their parents and higher resistance to stripe rust (Zhang et al. 2022). The molecular markers in this study were derived from SSR markers *Xgwm192* and *Xgwm251* discovered by Lu et al. (2014). They were closely linked to *Yr62* with genetic distances of 3.3 cM and 2.0 cM, respectively. The polymorphism of these markers was 100% in 140 wheat cultivars in the United States. In this study, after *Yr62* was introduced into four major cultivars in China, the descendant lines containing *Yr62* still maintained stable disease resistance. Several lines detected without *Yr62* in the four cross combinations still have a high level of resistance to stripe rust. The first possible reason is that the genetic distance between the SSR marker used in this study and *Yr62* is more than 1.0 cM, leading to recombination between genes, so it is not detected (Gupta et al. 1999). The second possible reason is that the recipient parent contains unknown disease resistance genes, so the field phenotype of lines without *Yr62* maintains a similar resistance to stripe rust to the recipient parent. The third reason may be that the cross population used in this study is different from the *Yr62* mapping population, so there is a different recombination between molecular markers and genes, resulting in a few lines without the *Yr62* gene detected by molecular markers. To sum up, the number of lines with *Yr62* detected by molecular markers is less than the actual number. Due to the above possibility, *Yr62* has not been detected in some progeny lines, so it is necessary to develop markers close to *Yr62* for marker-assisted selection breeding.

Evaluation of availability of disease-resistant germplasm resources

PI 660,060 (AvS/PI 192,252 F_4-103), which was previously developed from AvS/PI 192,252, was registered as a single *Yr62* gene reference line (Wang et al. 2012b). *Yr62* was susceptible to both low-temperature (4–20 °C) and high-temperature (10–30 °C) greenhouse seedling tests (IT 8), while it showed high resistance (IT 1–2) from the elongation stage in field tests abroad (Lu et al. 2014). In China, *Yr62* is susceptible to CYR32, CYR33, and PST-V26 at seedling stage (IT 8), and highly resistant to CYR32, CYR33, and PST-V26 at elongation stage (IT 2–3, DS 2–30%) at Yangling, Shaanxi and Tianshui, Gansu field test sites (Zhou et al. 2015). In this study, *Yr62* showed high resistance to stripe rust (IT 1, 1, 1, 0; DS 10, 15, 10, 0) at the adult plant stage from 2018 to 2022. In 2022, the IT of *Yr62* was 0, which may be because race virulence in 2022 is not very strong, so the *Yr62* phenotype is immune. All this proves that *Yr62* has a high level of durable HTAP resistance to stripe rust. Therefore, *Yr62* was selected to be introduced into Chinese main cultivars in this study to provide excellent resistant germplasm resources for China.

In wheat stripe rust epidemic years, cultivars with different levels of resistance were often associated with yield losses ranging from 10 to 70% and affected wheat quality (Afzal et al. 2007; Chen 2005a; Zhou et al. 2021). Therefore, one of the main concerns of breeders in practical production is to improve the availability of resistance origin while taking into account yield, quality, and the ability to combine disease resistance (Deng et al. 2017; Figueroa et al. 2018; Nelson et al. 2018). In this study, nine lines were selected based on disease resistance and agronomic traits between donor and recipient parents. In the breeding process, SWUST-441 (IT 1, DS 5) and SWUST-447 (IT 1, DS 5) were found to have the same good resistance compared to the donor resistant parent PI 660,060. However, the plant height for both lines was over 105 cm; SWUST-442 (IT 2, DS 10) and SWUST-421 (IT 3, DS 30) had good disease resistance to stripe rust, but the TGW was less than the selected lines. However, SWUST-441, SWUST-447, SWUST-442, and SWUST-421 were not selected due to these disadvantages. The final nine wheat lines selected were not only highly resistant to wheat stripe rust, but also superior to the donor resistant parent

PI 660,060 in important agronomic traits such as PH, SL, TN, and SGN. These selected lines could be applied in further practical production. Heritability constructs of breed traits provide a reference index for breeders in population construction and selection of progeny traits. In this study, PH, SN, TGW, and SL of the line with the H6172 background were higher than those of the other crosses, while TN of the line with the ZM9023 background was higher than that of the other crosses.

In this study, recipient parents LX987, AK58, ZM9023, and H6172 were used as materials, all of which were released cultivars with good agronomic characteristics and yield potential. Through the evaluation of the resistance, agronomic traits, and yield of the Recombinant Inbred Lines (RIL) of the four crosses, the results showed that the resistance gene *Yr62* was successfully transferred to the different genetic backgrounds, and growth and development status of the grain was better than that of the recipient parents. Nevertheless, the present study showed that line *Yr62* can be used as a useful stripe rust resistance germplasm resource for developing wheat cultivars with durable resistance.

Conclusion

In this study, *Yr62* resistance gene of wheat stripe rust resistance gene PI 660,060 was transferred to the main domestic wheat cultivars, such as LX987, AK58, ZM9023, and H6172, by combining conventional breeding and molecular marker identification. Through molecular marker detection and agronomic trait evaluation, progeny lines with the high level of stripe rust resistance and excellent agronomic traits were selected from the four crosses, which can be used to cultivate multi-gene polymerized cultivars and have important application value and research value. Practice shows the advantages of this method, saving a lot of human and material resources, and finally achieving the purpose of selecting excellent new wheat cultivars.

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Author contribution ZJN and ZXC tested markers, analyzed data, and wrote the first draft of the manuscript. ZX, WYQ, and MCH conducted field testing and collected phenotypic data. ZJN, TWJ, and TR evaluated the population and parents. YSZ and ZXL selected the lines of population. CXM, LX, XCJ, and ZXL contributed suggestions for the project and revised the first draft. CXM, KZS, and ZXL conceived the project and produced the final version of the manuscript. All authors provided suggestions during revision of the manuscript.

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Data availability Data can be accessed by contacting the first author and corresponding author (eli6951@sina.com).

Declarations

Conflict of interest The authors declare no competing interests.

References

Afzal S, Afzal U-H, Ahmedani MS, Bashir S, Ur A, Rattu R (2007) Assessment of yield losses caused by Puccinia striiformis triggering stripe rust in the most common wheat varieties. Pak J Bot 39:2127–2134. https://doi.org/10.1094/MPMI-20-12-1604

Akbari M, Wenzl P, Caig V, Carling J, Xia L, Yang S, Uszynski G, Mohler V, Lehmensiek A, Kuchel H, Hayden MJ, Howes N, Sharp P, Vaughan P, Rathmell B, Huttner E, Kilian A (2006) Diversity arrays technology (DArT) for high-throughput profiling of the hexaploid wheat genome. Theor Appl Genet 113(8):1409–1420. https://doi.org/10.1007/s00122-006-0365-4

Asseng S, Foster I, Turner NC (2011) The impact of temperature variability on wheat yields 17:997-1012. https://doi.org/10.1111/j.1365-2486.2010.02262.x

Brown JK (2015) Durable resistance of crops to disease: a Darwinian perspective. Annu Rev Phytopathol 53:513–539. https://doi.org/10.1146/annurev-phyto-102313-045914

Chakraborty S, Newton AC (2011) Climate change, plant diseases and food security: an overview. 60:2–14. https://doi.org/10.1111/j.1365-3059.2010.02411.x

Chen XM (2005a) Epidemiology and control of stripe rust [Puccinia striiformis f. sp tritici] on wheat. Can J Plant Path 27:314–337. https://doi.org/10.1080/07060660509507230

Chen XM (2005b) Epidemiology and control of stripe rust [Puccinia striiformis f. sp. tritici] on wheat. Can J Plant Path 27:314–337. https://doi.org/10.1080/07060660509507230

Chen XM (2013) Review article: High-temperature adult-plant resistance, key for sustainable control of stripe rust. Am J Plant Sci 04:608–627. https://doi.org/10.4236/ajps.2013.43080

Chen XM, Line RF, Leung H (1998) Genome scanning for resistance-gene analogs in rice, barley, and wheat by high-resolution electrophoresis. Theor Appl Genet 97:345–355. https://doi.org/10.1007/s001220050905

Christiansen MJ, Feenstra B, Skovgaard IM, Andersen SB (2006) Genetic analysis of resistance to yellow rust in hexaploid wheat using a mixture model for multiple crosses. Theor Appl Genet 112:581–591. https://doi.org/10.1007/s00122-005-0128-7

Collard BCY, Jahufer MZZ, Brouwer JB, Pang ECK (2005) An introduction to markers, quantitative trait loci (QTL) mapping and marker-assisted selection for crop improvement: the basic concepts. Euphytica 142:169–196. https://doi.org/10.1007/s10681-005-1681-5

Cuc L, Mace E, Crouch J, Quang V, Long T, Varshney R (2008) Isolation and characterization of novel microsatellite markers and their application for diversity assessment in cultivated groundnut (Arachis hypogaea). BMC Plant Biol 8:55. https://doi.org/10.1186/1471-2229-8-55

Deng Y, Zhai K, Xie Z, Yang D, Zhu X, Liu J, Wang X, Qin P, Yang Y, Zhang G, Li Q, Zhang J, Wu S, Milazzo J, Mao B, Wang E, Xie H, Tharreau D, He Z (2017) Epigenetic regulation of antagonistic receptors confers rice blast resistance with yield balance. Science 355(6328):962–965. https://doi.org/10.1126/science.aai8898

Facundo T, Stephen P, Barneix AJ (2017) Breeding for increased grain protein and micronutrient content in wheat: ten years of the *GPC-B1* gene. J Cereal Sci 73:183–191. https://doi.org/10.1016/j.jcs.2017.01.003

Figueroa M, Hammond-Kosack KE, Solomon PS (2018) A review of wheat diseases–a field perspective. Mol Plant Pathol 19:1523–1536. https://doi.org/10.1111/mpp.12618

Gao L, Yan X, Li X, Guo G, Hu Y, Ma W, Yan Y, (2011) Proteome analysis of wheat leaf under salt stress by two-dimensional difference gel electrophoresis (2D-DIGE). https://doi.org/10.1016/j.phytochem.2010.12.008

Guo SB, Chen JH, Shi HZ, Kang ZS, Huang LL (2010) Discussion on control strategies and methods of wheat stripe rust. Hubei Agric Sci 972–975. https://doi.org/10.14088/j.cnki.issn0439-8114.2010.04.023

Gupta PK, Varshney RK, Sharma PC, Ramesh BJPB (1999) Molecular markers and their applications in wheat breeding. Plant Breeding 118:369–390. https://doi.org/10.1046/j.1439-0523.1999.00401.x

Juroszek P, von Tiedemann A (2013) Climate change and potential future risks through wheat diseases: a review. Eur J Plant Pathol 136:21–33. https://doi.org/10.1007/s10658-012-0144-9

Kang ZS, Wang X-J, Zhao J, Tang C-L, Huang L-L (2015) Advances in research of pathogenicity and virulence variation of the wheat stripe rust fungus Puccinia striiformis f. sp. tritici. Chin Agric Sci 48(17):3439–3453. https://doi.org/10.3864/j.issn.0578-1752.2015.17.011

Khanfri S, Boulif M, Lahlali RJNSB (2018) Yellow rust (Puccinia striiformis): a serious threat to wheat production worldwide. https://doi.org/10.15835/NSB10310287

Li W, Song GQ, Li JH, Li YL, Zhan SJ, Zhang RZ, Gao J, Gu TT, Li GY (2020) Molecular detection of four pleiotropic disease resistance genes in wheat. Mailei Acta Agron Sin 40:395–400

Li ZQ, Wan MN, Jia MG, Lu HP, Shan HS, Kang ZS (1997) Study on the epidemiology and control strategy of wheat stripe rust in South Gansu. Acta Agriculture Boreali-Occidentalis Sinica 02:10–14

Line RF (2002) Stripe rust of wheat and barley in North America: a retrospective historical review. Annu Rev Phytopathol 40:75–118. https://doi.org/10.1146/annurev.phyto.40.020102.111645

Line RF, Chen X (1995) Successes in breeding for and managing durable resistance to wheat rusts. Plant Disease 79:1254–1255

Liu B, Liu T, Zhang Z, Jia Q, Wang B, Gao L, Peng Y, Jin S, Chen W (2017) Discovery and pathogenicity of CYR34, a new race of Puccinia striiformis f.sp.tritici in China. Acta Phytopathol Sin 681–687. https://doi.org/10.13926/j.cnki.apps.000071

Liu K, Shen L (2002) Features of the epidemic of wheat strip rust in 2002 in Sichuan and the causes underlying it. J Southwest Agric Univ, 299–302

Liu T, Fedak G, Zhang L, Zhou R, Chi D, Fetch T, Hiebert C, Chen W, Liu B, Liu D, Zhang H, Zhang BA-O (2020) Molecular marker based design for breeding wheat lines with multiple resistance and superior quality. Plant Dis 104(10):2658–2664. https://doi.org/10.1094/PDIS-02-20-0420-RE

Lowe I, Cantu D, Dubcovsky J (2011) Durable resistance to the wheat rusts: integrating systems biology and traditional phenotype-based research methods to guide the deployment of resistance genes. https://doi.org/10.1007/s10681-010-0311-z

Lu Y, Wang MN, Chen XM, See D, Chao SM, Jing JX (2014) Mapping of *Yr62* and a small-effect QTL for high-temperature adult-plant resistance to stripe rust in spring wheat PI 192252. Theor Appl Genet 127:1449–1459. https://doi.org/10.1007/s00122-014-2312-0

Ma KJ, Li XY, Li Y, Wang ZH, Zhao BG, Li Q (2021) Disease resistance and genes in 146 wheat cultivars (lines) from the Huang-Huai-Hai region of China. Agronomy 11:1025. https://doi.org/10.3390/agronomy11061025

Meng L, Li H, Zhang L, Wang J (2015) QTL IciMapping: integrated software for genetic linkage map construction and quantitative trait locus mapping in biparental populations. The Crop Journal 3:269–283. https://doi.org/10.1016/j.cj.2015.01.001

Nelson R, Wiesner-Hanks T, Wisser R, Balint-Kurti P (2018) Navigating complexity to breed disease-resistant crops. Nat Rev Genet 19:21–33. https://doi.org/10.1038/nrg.2017.82

Niks RE, Qi X, Marcel TC (2015) Quantitative resistance to biotrophic filamentous plant pathogens: concepts, misconceptions, and mechanisms. Annual review of phytopathology 53. https://doi.org/10.1146/annurev-phyto-080614-115928

Parlevliet JE (1985) 16 - Resistance of the non-race-specific type. In: Roelfs AP and Bushnell WR (eds.), Diseases, distribution, epidemiology, and control. Academic Press. https://doi.org/10.1016/B978-0-12-148402-6.50024-9

Rasheed A, Mujeeb-Kazi A, Ogbonnaya FC, He Z, S R. (2018) Wheat genetic resources in the post-genomics era: promise and challenges. Ann Bot 121:603–616. https://doi.org/10.1093/aob/mcx148

Roelfs AP, Bushnell WR (1985) Preface. In: Roelfs AP and Bushnell WR (eds.), Diseases, distribution, epidemiology, and control. Academic Press. https://doi.org/10.1016/B978-0-12-148402-6.50007-9

Savadi S (2018) Molecular regulation of seed development and strategies for engineering seed size in crop plants. Plant Growth Regul 84:401–422. https://doi.org/10.1007/s10725-017-0355-3

Sun QD, Cao J, Ma YA, Chen DM, Liu BH (2004) Seed coating of wheat variety HM6172 and micro-fertilizer quantity test research. China Seed Industry 08:28–29. https://doi.org/10.19462/j.cnki.1671-895x.2004.08.014

Susanne Sukumaran S, Guzmán C, He X, Lan C, Bonnett D, Crossa J (2016) Molecular marker-based selection tools in spring bread wheat improvement: CIMMYT experience and prospects. In: R. S. Rajpal V, Raina S (ed.), Molecular breeding for sustainable crop improvement. Sustainable Development and Biodiversity, vol 11. Springer, Cham. https://doi.org/10.1007/978-3-319-27090-6_16

Wang G, Hu TZ, Li XJ, Dong N, Feng SW, Li G, Zhang LL, Ru ZG (2012) Detection of dwarfing genes in wheat variety AK58 and its parents. J Henan Agric Sci 41(9):22–25. https://doi.org/10.15933/j.cnki.1004-3268.2012.09.019

Wang J (2009) Inclusive composite interval mapping of quantitative trait genes 35:239–245. https://doi.org/10.3724/SP.J.1006.2009.00239

Wang M, Chen X, Xu L, Cheng P, Bockelman HE (2012b) Registration of 70 common spring wheat germplasm lines resistant to stripe rust. J Plant Regist 6:104–110. https://doi.org/10.3198/jpr2011.05.0261crg

Wang ZW, Xie XM, Ma WX, Xie TX, Chen RX (2010) Research on the character of winter wheat—Lunxuan 987 growth and development and its water-saving and high-yielding cultivation technique. Anhui Agric Sci 38(23):12394–12396. https://doi.org/10.13989/j.cnki

Wellings CR (2011) Global status of stripe rust: a review of historical and current threats. Euphytica 179:129–141. https://doi.org/10.1007/s10681-011-0360-y

William HM, Singh RP, Huerta-Espino J, Palacios G, Suenaga K, (2006) Characterization of genetic loci conferring adult plant resistance to leaf rust and stripe rust in spring wheat. https://doi.org/10.1139/g06-052

Wu P, Xie J, Hu J, Qiu D, Liu Z, Li J, Li M, Zhang H, Yang L, Liu H, Zhou Y, Zhang Z, Li H (2018) Development of molecular markers linked to powdery mildew resistance gene *Pm4b* by combining SNP discovery from transcriptome sequencing data with bulked segregant analysis (BSR-Seq) in wheat. https://doi.org/10.3389/fpls.2018.00095

Xiang C, Feng JY, Wang MN, Chen XM, See DR, Wan AM, Wang T (2016) Molecular mapping of stripe rust resistance gene Yr76 in winter club wheat cultivar Tyee. https://doi.org/10.1094/PHYTO-01-16-0045-FI

Xiang YJ, Zhang ZY, Shen L, Kuang WJ, Ji HL, Ni JY, Chen WQ, Peng YL (2013) Virulence Dynamics of Puccinia striiformis f. sp. tritici in Sichuan Province from 2005 to 2010. Southwest China J Agric Sci 26:1858–1863. https://doi.org/10.16213/j.cnki.scjas.2013.05.016

Xue W, Xu X, Mu JM, Wang QL, Wu JH, Huang LL, Kang ZS, Huang LL, Han DJ (2014) Evaluation of strip rust resistance and genes in Chinese elite wheat varieties. J Wheat Crops 34:1054–1060. https://doi.org/10.7606/j.issn.1009-1041.2014.08.06

Yang L (2006) Breeding and popularization of wheat cultivar LX 987. Dissertation, Chinese Academy of Agricultural Sciences

Zadoks JC, Chang TT, Konzak CF (1974) A decimal code for the growth stages of cereals. Weed Res 14:415–421. https://doi.org/10.1111/j.1365-3180.1974.tb01084.x

Zeng QD, Zhao J, Wu JH, Zhan GM, Han DJ, Kang ZS (2022) Wheat strip rust and integration of sustainable control strategies in China 9:37–51. https://doi.org/10.15302/j-fase-2021405

Zhang M, Fang TH, Zhou XL, Chen XM, Li X, Feng JY, Yang SZ, Kang ZS (2022) Combination of marker-assisted backcross selection of *Yr59* and phenotypic selection to improve stripe rust resistance and agronomic performance in four elite wheat cultivars. Agronomy 12:497. https://doi.org/10.3390/agronomy12020497

Zhou XL, Fang TH, Li KX, Huang KB, Ma CH, Zhang M, Li X, Yang SZ, Ren RS, Zhang PP (2021) Yield losses associated with different levels of stripe rust resistance of commercial wheat cultivars in China. Phytopathology® 112:1244–1254. https://doi.org/10.1094/PHYTO-07-21-0286-R

Zhou XL, Han DJ, Chen XM, Gou HL, Guo SJ, Rong L, Wang QL, Huang LL, Kang ZS (2014) Characterization and molecular mapping of stripe rust resistance gene *Yr61* in winter wheat cultivar Pindong 34. Theor Appl Genet 127(11):2349–2358. https://doi.org/10.1007/s00122-014-2381-0

Zhou XL, Zhan GM, Huang LL, Han DJ, Kang ZS (2015) Evaluation of resistance to stripe rust in eighty abroad spring wheat germplasms. Sci Agric Sinica 48:1518–1526. https://doi.org/10.3864/j.issn.0578-1752.2015.08.06

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